hedged_blend

portfolio strategy | family: portfolio | status: validated | derived from `calm_weighted`

What this strategy does

Market-neutral inverse-vol equity book (0.8x) + corrected-quote Kelly options book (1.6x), on one NAV

Why it should work. The sleeves are nearly uncorrelated with each other and with the market, so a blend can diversify. After the fixes the options sleeve earns about nothing, so the blend is essentially its equity sleeve (see docs/STRATEGIES.md section 5). Weights (set before the fixes, not re-optimised) gave each sleeve equal stand-alone risk at ~5% portfolio volatility.

Design notes (from the strategy's source file).

PortfolioStrategy: a fixed-weight, daily-rebalanced blend of other registered strategies.

A sleeve weight is a multiple of fund NAV allocated to that strategy's own sizing: weight 1.0 runs the strategy exactly as backtested on the whole NAV, 0.8 runs it 20% smaller. Weights need not sum to one -- the sleeves are low-correlation books that each use only part of their capital (the options sleeve spends at most its premium budget, the equity sleeve is margin-financed long/short), so they can share one NAV. Returns compound daily: r_p = sum(w_i r_i), NAV_t = NAV_{t-1} (1 + r_p).

Sleeve results come from the strategy registry itself (re-run in-process, ~seconds), so the blend always reflects the current sleeve definitions. Days with no sleeve activity (options in 2011, where the source file is corrupt) contribute zero for that sleeve.

parameter	value
w_calm_weighted	0.8
w_options_cap20	1.6

Performance

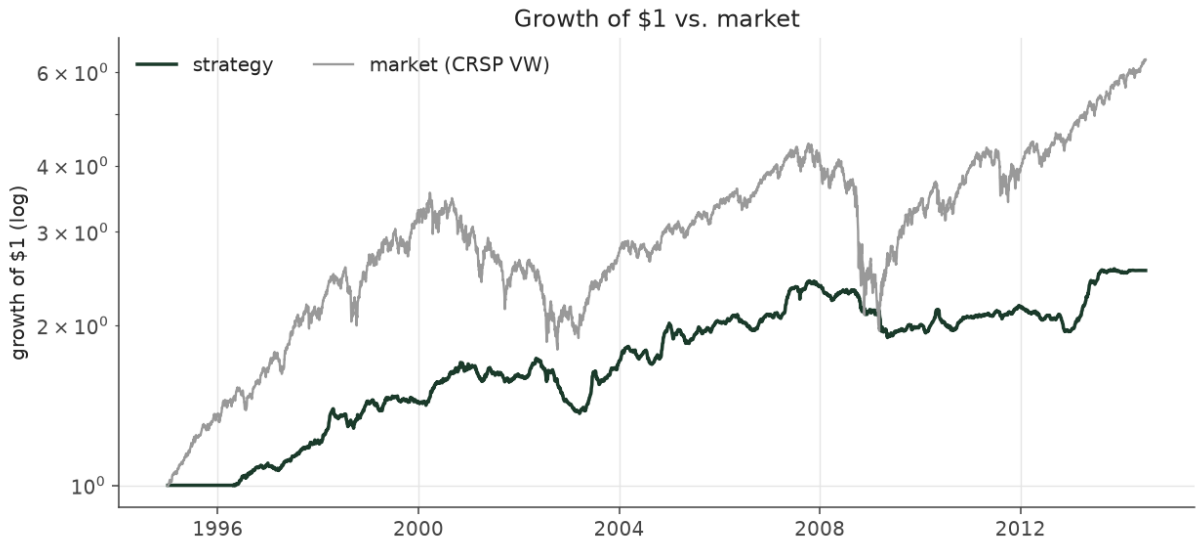
metric	hedged blend	market (CRSP VW)
Annualized return	4.90%	9.94%
Annualized volatility	4.46%	19.35%
Sharpe (NAV-pct)	1.09	0.59
Sortino	1.61	
Max drawdown	-21.8%	-55.5%
Calmar	0.22	
Final NAV ($10M start)	$25,382,788	
Worst day	-2.93%	
% positive days	50.8%	
Sharpe 1996-2001	1.82	
Sharpe 2002-2007	1.31	
Sharpe 2008-2013	0.25	

Correlation of daily returns with the market: **0.08**.

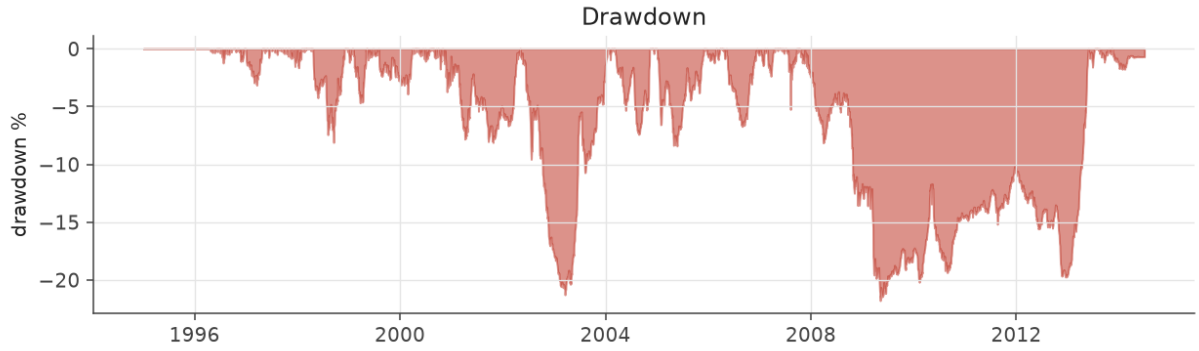
• best Sharpe among portfolio: `calm_blend` (1.28); this one: 1.09

• best return among portfolio: `calm_blend` (0.058); this one: 0.049

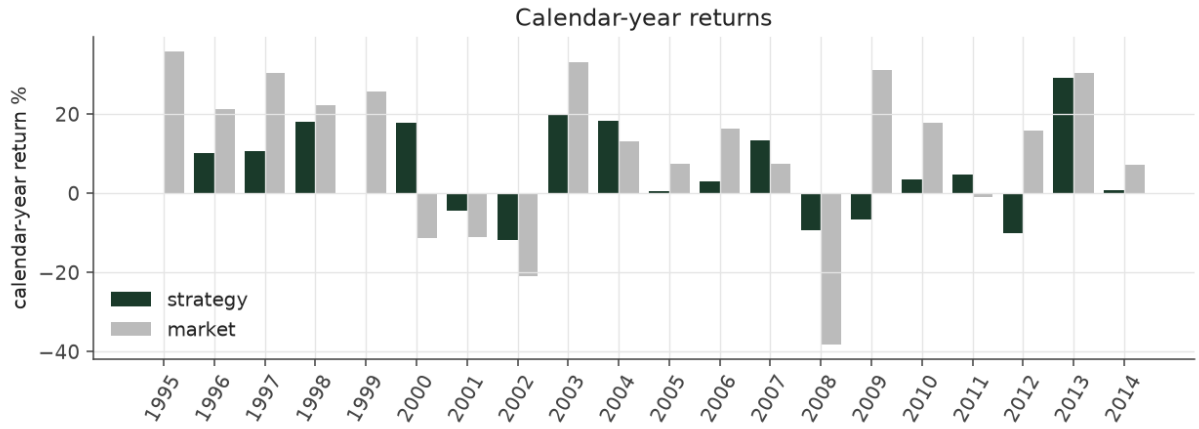
• best Calmar among portfolio: `calm_blend` (0.288); this one: 0.225



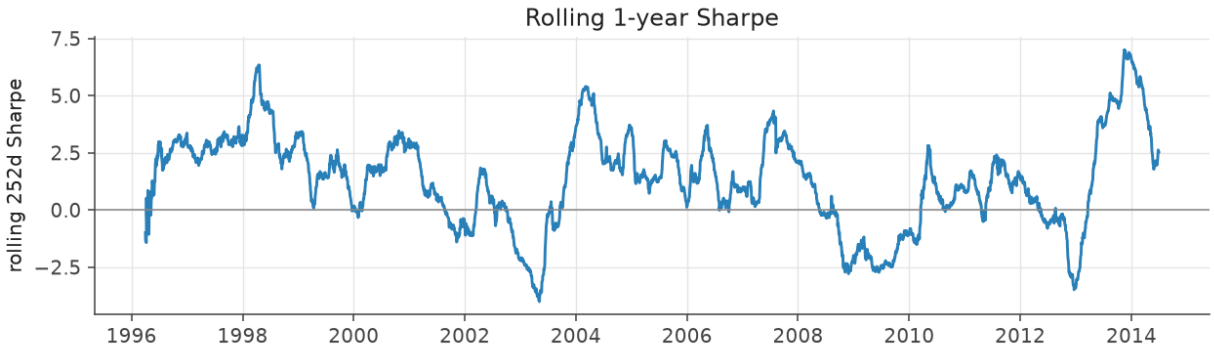
NAV (log scale)



Drawdown from running peak



Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Each sleeve's contribution to the blend's cumulative P&L

sleeve	weight	pnl M	share of pnl
calm_weighted	0.80	14.28	92.8%
options_cap20	1.60	1.10	7.2%

• Correlation between the sleeves' daily returns: **-0.06**.

• Blend market beta 0.018, correlation to the market 0.08.

Return, volatility and Sharpe of the blend are in the Performance table above; the cross-strategy report (`python -m pead compare`) shows the full weight frontier, the walk-forward (out-of-sample) weights and the volatility-target table.